Section A

Answer **all** the questions in this section.

Question 1

Fig. 1.1 shows the results of an experiment in which samples containing the same concentration of enzyme and substrate were kept at different temperatures for periods of one, two and five hours. The quantities of product formed were then determined.

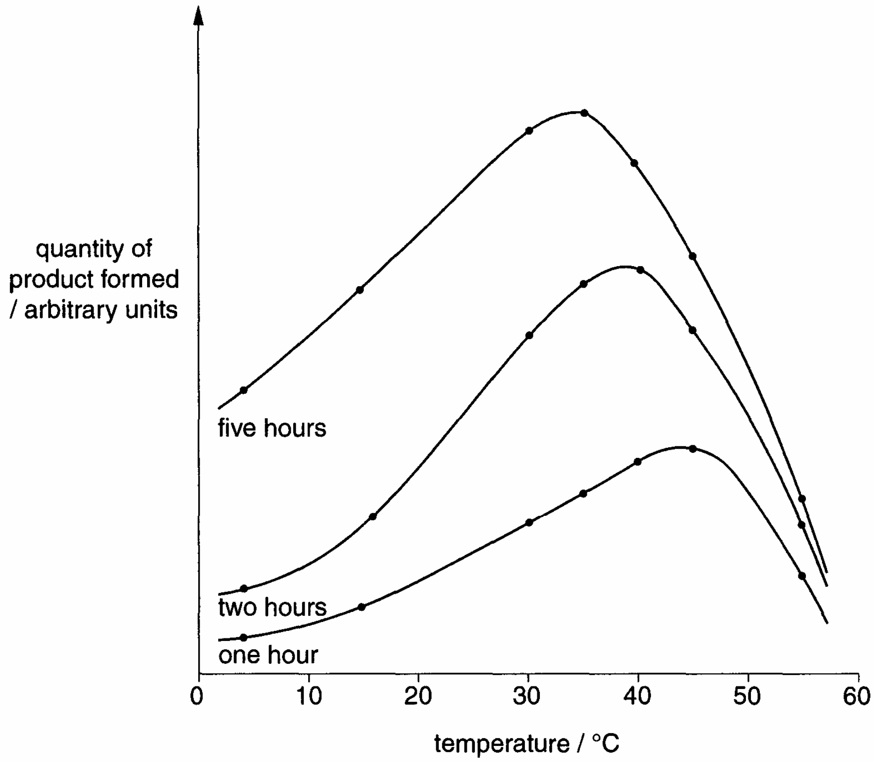

Fig. 1.1

(a) Explain the effect of increasing temperature on the quantity of product formed for the samples kept at different temperatures for one hour. [4]

(b) Explain why the optimum temperature is lower if the quantity of product formed is measured after five hours rather than one hour. [2]

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Fig. 1.2 shows the effect of increasing substrate concentration on the rate of an enzyme-catalysed reaction. All other variables were controlled.

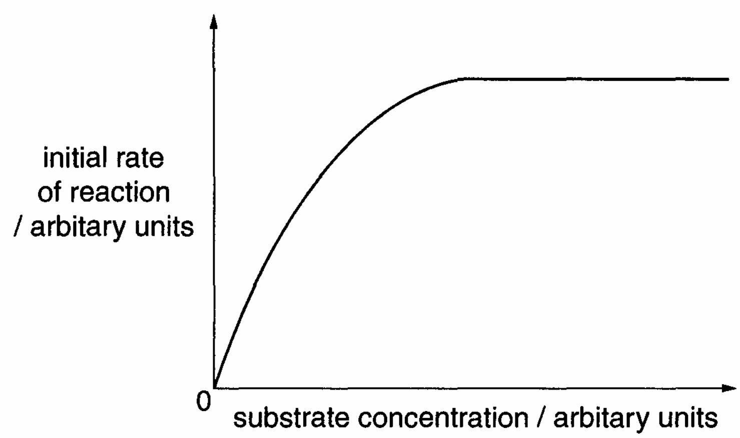

Fig. 1.2

(c) Explain why an increase in substrate concentration at low substrate concentrations increases the initial rate of reaction but an increase at high substrate concentrations does not do so. [3]

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(d) State what change in conditions would cause an increase in the initial rate of reaction at high substrate concentrations. [1]

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[Total: 10]

Question 2

Pure-breeding pea plants with round, yellow seeds were crossed with pure-breeding pea plants with wrinkled, green seeds. The offspring all had round, yellow seeds. These seeds were grown and the resultant plants allowed to self-fertilise.

This produced 1112 offspring with the following characteristics.

630	round, yellow seeds
202	round, green seeds
216	wrinkled, yellow seeds
64	wrinkled, green seeds

(a) Using the symbols **R** for round, **r** for wrinkled, **B** for yellow and **b** for green, draw a genetic diagram to explain these results. [4]

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(b) With reference to **(a)**, explain why the wrinkled, green seeds offspring produced are only pure-breeding, while the round, yellow seeds did not. [3]

A ratio of 9:3:3:1 was expected. A chi-squared test was carried out to test the significance of the differences between the observed and expected results. This gave a value of 0.47.

Table 2.1

Probability	0.99	0.98	0.95	0.90	0.50	0.10	0.05	0.02	0.01
At 3 degrees of freedom	0.12	0.19	0.35	0.58	2.4	6.3	7.8	9.8	11.3

(c) With reference to the Table 2.1, explain how the value for the chi-squared test supports the hypothesis that these are two pairs of segregating alleles at two loci. [2]

[Total: 9]

Question 3

In an investigation of aerobic respiration, isolated mitochondria were added to a prepared medium containing succinate and inorganic phosphate. Succinate is a 4-carbon compound, which occurs in the Krebs cycle, and can be used as a respiratory substrate. The medium was saturated with oxygen. Equal amounts of ADP were added at one-minute intervals, and measurements were taken from the oxygen concentration in the medium. Fig. 3.1 graph shows the results.

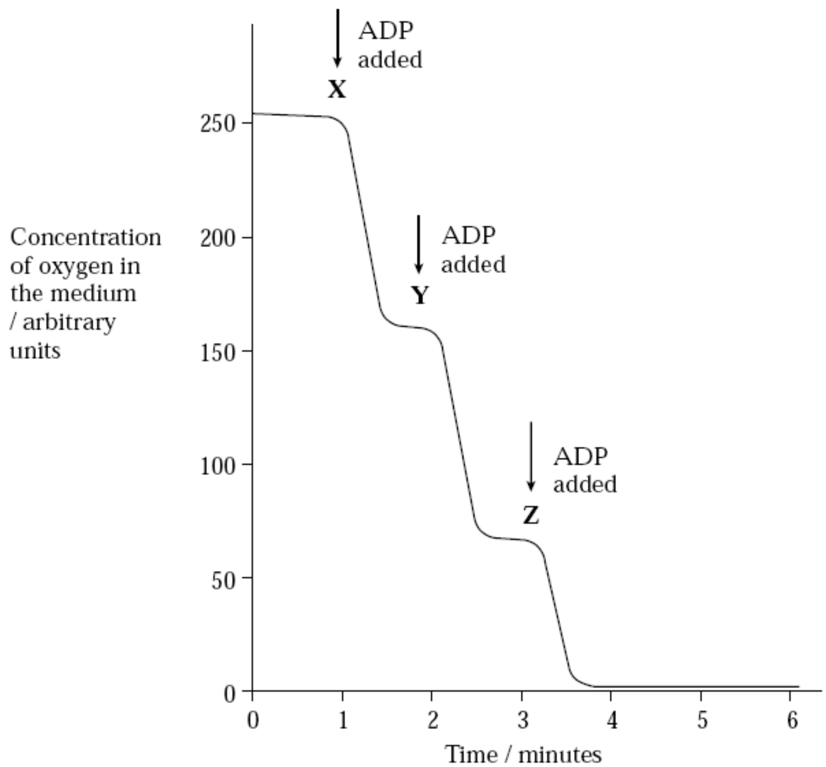

Fig. 3.1

(a) Why was inorganic phosphate added to the medium? [1]

(b) Explain why the oxygen concentration in the medium decreased after adding ADP at **X**. [3]

For Examiner's Use

(c) State why the fall in oxygen concentration, following the addition of ADP, was less at **Z** than at **Y**. [1]

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Fresh mitochondria were isolated from cells and a similar experiment was carried out. This time the medium contained glucose instead of succinate. Again, the medium was saturated with oxygen, and excess ADP was added. However, there was almost no fall in oxygen concentration, even after 10 minutes.

(d) Suggest and explain a reason for this observation. [2]

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[Total: 7]

Question 4

Bacteria can be genetically modified to produce insulin for human use. To achieve this, human insulin genes are transferred into bacteria. Plasmids containing two antibiotic resistance genes, one coding for resistance to tetracycline and one for resistance to ampicillin, are used to carry out this transfer. A restriction enzyme was used to cut up the human DNA and plasmids. Fig. 4.1 shows the different fragments of human DNA and the type of cut plasmid that was produced.

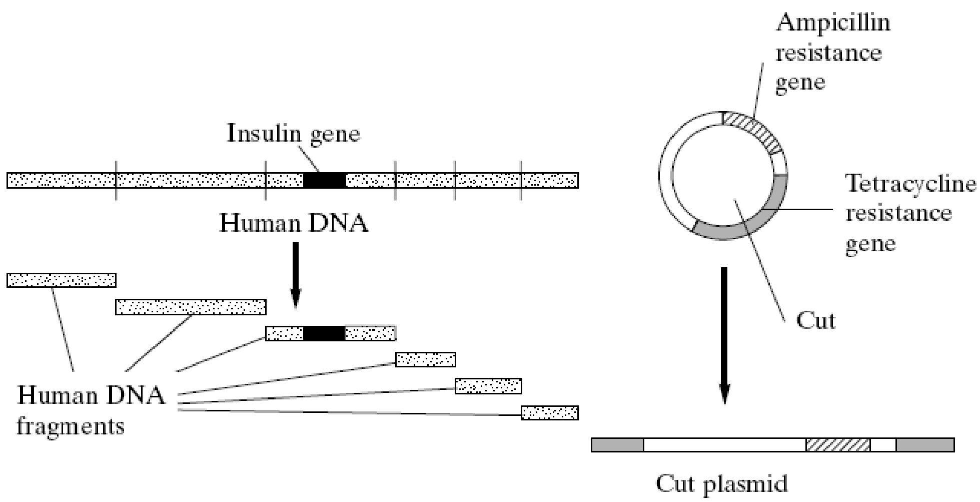

Fig. 4.1

(a) Describe the main structural features of a plasmid. [2]

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(b) Suggest why the restriction enzyme has cut the human DNA in many places but has cut the plasmid DNA only once. [2]

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The fragments of human DNA and the cut plasmids were mixed together with DNA ligase. Several types of plasmid were formed. Some contained human DNA in the centre of the gene coding for resistance to tetracycline. The different types of plasmid are shown in Fig. 4.2.

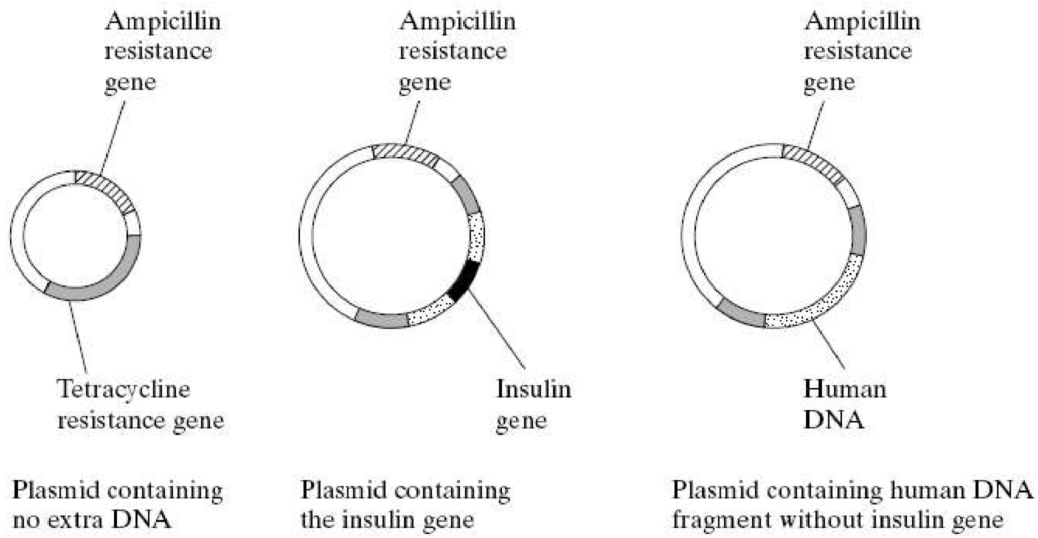

Fig. 4.2

(c) Explain what causes several types of plasmid to be formed. [2]

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The plasmids were mixed with the bacteria. Some bacteria took up the plasmids.

(d) Explain how it is possible to distinguish between bacteria which have taken up a plasmid with human DNA and those which have taken up a plasmid without any extra DNA. [5]

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(e) Describe how the bacterium containing the insulin gene is used to obtain sufficient insulin for commercial use. [3]

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[Total: 14]

Section B

Answer **ONE** question.

Write your answers on the separate writing papers provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 5

(a) Discuss the role of mRNA and tRNA in protein synthesis. [6]

(b) Describe how control elements influence transcription in eukaryotic cells. [8]

(c) Describe the eukaryotic processing of pre-mRNA. [6]

[Total: 20]

Question 6

(a) Describe how (polymerase chain reaction) PCR is used to clone DNA. [6]

(b) Legislation in many countries prohibits the import and export of endangered species.

Explain the process of genetic fingerprinting and suggest how it can be used to verify whether an organism belonging to an endangered species has been bred in captivity. [6]

(c) Describe how genomic DNA and cDNA libraries are produced. [8]

[Total: 20]

End of H1 Paper 2